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Review Article

Effects of excessive video game and social networking use on mental well-being and academic performance of university students: A systematic analysis

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A R T I C L E I N F O

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A B S T R A C T

This study examines the impact of excessive video games and social network use on university students' physical activity, mental well-being, and academic performance. This systematic review of 16 empirical studies, conducted in accordance with the PRISMA protocol, highlights the complex relationship between excessive digital use and its negative effects, including decreased academic performance and adverse impacts on both mental and physical health. Findings confirm links between information overload, social media addiction, academic stress, procrastination, and mental health issues like anxiety and depression, especially in women. Prolonged video gaming is also associated with sedentary behaviour and health issues, such as headaches and fatigue. Although most studies highlight adverse effects, emerging evidence suggests that the usage context can moderate these outcomes, underscoring the need for tailored interventions. Educational programs and public health strategies are essential for promoting balanced technology use and improving the physical well-being and academic outcomes of university students.

1. Introduction

The concept of mental well-being is a multidimensional construct that includes emotional stability, absence of anxiety or depression, and positive social functioning (World Health Organization, 2022). Academic performance (AP) refers to students' measurable learning outcomes, typically evaluated through Grade Point Average (GPA) or academic progress indicators (York et al., 2015; Rodríguez-Fernández et al., 2020). Given that well-being and academic performance are closely interrelated, it is crucial to analyse them together when studying the effects of excessive digital consumption among university students (Pérez-Jorge, Boutaba-Alehyan, González-Contreras, & Pérez-Pérez, 2025). The conceptualization of the internet and technology-related addictions has evolved significantly over the past three decades. Kimberly Young (1998) was among the first to define internet addiction as a clinical disorder, laying the groundwork for a growing body of empirical studies.

The conceptualization of technology-related behavioral addictions has evolved significantly in recent years. The DSM-5 introduced Internet Gaming Disorder as a condition for further clinical study, and the WHO later categorized Gaming Disorder as a mental health condition, reflecting growing concern about excessive digital engagement. The COVID-19 pandemic intensified the reliance on digital platforms for academic, social, and recreational purposes, increasing the potential for problematic and compulsive use among university students (Pérez-Jorge et al., 2020). However, excessive use does not arise solely from external circumstances; it also relates to motivational, emotional, and social factors that influence coping and identity formation in young adults. Within this context, understanding the relationship between excessive digital use and academic performance becomes essential for developing meaningful preventive and educational strategies.

The COVID-19 pandemic further accelerated these patterns by reinforcing reliance on digital platforms for academic, social, and entertainment purposes, thereby increasing the risk of problematic use. It is important to note that, although changes in digital usage patterns have been observed during the COVID-19 pandemic, this analysis does not seek to establish a direct causal relationship. The mention of COVID is limited to contextualizing potential factors that may have influenced university students' digital habits. More recently, the emergence of short-form video platforms like TikTok, the integration of AI-powered

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algorithms to maximize user engagement, and the growing popularity of immersive gaming experiences in virtual or augmented reality (VR/AR) raise new questions about the impact of evolving digital environments on university students' mental health and academic outcomes (Ariño-Mateo et al., 2024). This study examines excessive consumption of video games and social networks among university students and their impact on AP and well-being. Numerous studies have warned about the consequences of abusive video game use in students.

Similarly, the problematic social media (SM) use has raised concerns. growing concern due to its psychological and academic implications. While much of the existing literature has focused separately on academic or psychological outcomes, this review seeks to bridge that gap by examining both dimensions in tandem. Addressing these variables simultaneously provides a more comprehensive understanding of how digital habits affect university students holistically. Despite existing studies on digital overuse, few examine academic and psychological outcomes together. This review examines these outcomes jointly to provide a more comprehensive understanding of how excessive digital use affects university students holistically, highlighting the interconnections between mental well-being and academic performance. Therefore, this study aims to identify the combined effects of excessive use of video games and social networks on the psychological well-being and academic performance of university students.

The theoretical framework is structured into three thematic strands: AP, mental health impact, and the combined effects of excessive information and communication technology use. These trends highlight the urgency of adopting a forward-looking, integrative perspective in current research. A growing body of evidence has established a link between excessive digital media use. A growing body of evidence has established a link between excessive digital media use and both academic and mental health consequences, for instance recent studies have shown that excessive use of video games and social networking sites is associated with both decreased AP and increased symptoms of anxiety, depression, and stress among university students (Hussain & Griffiths, 2021; Zhang et al., 2022). Moreover, such behaviours have been identified as potential coping mechanisms for emotional difficulties, further aggravating their impact on psychological well-being (Kircaburun & Griffiths, 2018; Zhang et al., 2022). Taken together, these studies suggest that such behaviours may function as maladaptive coping mechanisms, exacerbating existing emotional difficulties (Pérez-Jorge, Boutaba-Alehyan, González-Contreras, & Pérez-Pérez, 2025).

Azizi et al. (2019) identified moderate levels of social media addiction (SMA) among students, which were more prevalent in males than in females and were linked to negative impacts on AP. They suggest university authorities provide workshops to inform students of SMA's adverse effects.

Alhusban et al. (2022), Pellegrino et al. (2022), and Noorunnahar et al. (2023) found a clear relationship between problematic SM use and poor AP. Alzahrani and Griffiths (2024) linked video game addiction to impaired AP due to reduced study time and increased fatigue. Meanwhile, Barreto-Cabrera et al. (2024) found that higher problematic video game use correlates with lower AP and poorer student decision-making. The Digital 2022 report by We Are Social and Hootsuite revealed a significant global increase in social media (SM) users, with 424 million new users added in the previous year, bringing the total number of SM users to approximately 58 % of the world's population. The national statistics provide crucial context for understanding the prevalence and normalization of social media use among young adults, particularly in Spain, where digital behaviors are deeply embedded in daily routines (Pérez-Jorge et al., 2024b). In Spain, the same report indicated that 87.1 % of the population—around 40.7 million people—were active SM users in 2022. On average, users spend 1 h and 53 min per day on these platforms, with WhatsApp (91 %) and Facebook (73.3 %) being the most frequently used. In Spain, 16-24-year-olds spend about 6 h daily on the internet, including 3 h on SM, with 80 % using WhatsApp and Facebook for 1 h and 53 min daily.

Dzul and Cauich (2021) examined SM xenophobia, finding that Facebook's xenophobic content was perceived as commonplace, even humorous, by young users, blurring the line between humor and discrimination. In the Canary Islands, mental health problems were reported as the highest nationwide (31 %), per Henares et al. (2020) and the National Institute of Statistics (INE) (2023).

The World Health Organization (2022) now considers video game addiction a mental disorder, with similar issues arising from SM and instant messaging addictions. Khoshakhlagh and y Faramarzi (2012) noted internet addiction's anxiety effects, and García del Castillo et al. (2019) linked SM use to maladjustment, depression, low self-esteem, mistrust, suicidal thoughts, and fear. A considerable proportion of students also used the internet for gambling, a concern as many were underage; 17 % of boys and 5 % of girls had gambled online. Varchetta et al. (2020) noted that increasing SMA levels heightened exclusion fears, and Fernández (2022) observed that social network use was most frequent among those with higher education levels (72 %). The IV "Plan Canario sobre Adicciones" aims to help students make autonomous decisions to avoid the use of abusive information and communication technologies, the internet, SM, and gambling ("Servicio de Coordinación Técnica de Atención a las Drogodependencias, 2020).

Moreno (2019) described Instagram as the most harmful social network for mental health. López (2022) similarly highlighted Instagram's effects on young women in the Canary Islands. Lozano-Blasco and Cortés-Pascual (2020) linked Facebook addiction with low self-esteem, depression, and social skills deficits, like Valencia-Ortiz et al. (2021), who showed SMA could lead to depression and social isolation. Young users struggled to distinguish between real and virtual environments, leading to poor AP scores and, in some cases, school dropouts. Alarcón-Allaín and Salas-Blas (2022) found that lower emotional intelligence among technical education students is negatively associated with psychological well-being and stability. Zhao (2021) noted that SMA negatively impacts the university students' subjective well-being, leading to dissatisfaction and depression. Jara, Leny, and Nicolás Calderón (2022) found that 98.5 % of medical students showed signs of SMA, with 66 % having moderate addiction levels. Suicide arguably constitutes the most visible consequence of mental health issues among youth, with video games and SM overuse contributing to social isolation, reduced motivation for interaction, and social skill deficits. This has led increasingly isolated youth to favor virtual interactions (Chamarro et al., 2024).

Chowdhury et al. (2024) found high anxiety, depression, and insomnia levels among Bangladeshi students, with risk factors like SMA, age, AP, smoking, and income patterns. Filiz and Kaya (2021) showed that as internet addiction and sleep disturbances rise, AP declines. Rahman et al. (2023) found that excessive SM time distracted students, affecting concentration and time management, leading to poor grades. Those with higher SMA suffered anxiety and stress, worsening AP. Aslan and Polat (2024) linked SM use for more than 7 h a day and visiting SM over 15 times daily to key addiction factors that negatively impact academic success. Although students with high academic self-efficacy could maintain a moderate GPA, SMA had an impact on their mental health. David et al. (2024) found that SMA induced mental disorders, like social anxiety, reduced academic engagement, affecting AP. Clark (2023), studying students in Thailand and China, found that while video games promoted socialization and skills like problem-solving, heavy use impaired AP and mental health.

González Batista (2022) noted high rates of suicidal behavior among youth. According to the "Fundación Española para la Prevención del Suicidio" (2022), 300 young Spaniards aged 15–29 committed suicide in 2020. The Canary Islands reported a significant increase in suicides from 2007 to 2020, holding the third-highest rate in Spain. In light of this concerning scenario, it is imperative to conduct research that integrates these dimensions to provide a holistic understanding of the phenomenon. Although numerous studies have examined the academic and psychological effects of excessive digital consumption, few have adopted a comprehensive framework that integrates both domains, thereby limiting understanding of how digital overuse simultaneously affects students' academic performance and mental well-being. There is a critical need to consolidate findings across these dimensions, particularly among university students, where mental health and AP are deeply interconnected. This review aims to address this gap by systematically analysing how excessive video games and SM use jointly affect student well-being and performance, offering evidence-based insights for educators, health professionals, and policymakers. Despite the growing number of studies addressing excessive use of video games and social networks among university students, these contributions are valuable but often remain limited in scope. No systematic review has integrated findings on AP, mental health, and physical well-being within a single synthesis.

Previous reviews have typically focused on single dimensions, such as academic outcomes (Rahman et al., 2023) or mental health risks (Kircaburun & Griffiths, 2018), without providing a comprehensive, multidimensional overview. Moreover, there is a lack of integrative reviews that consider diverse geographic regions and cultural contexts, particularly outside of North America and Western Europe. This gap highlights the urgent need for a systematic approach that summarizes existing knowledge and identifies cross-cutting patterns and emerging challenges. A systematic review, rather than a scoping or narrative review, was selected for its methodological rigour in evaluating the quality of empirical evidence and in synthesizing findings based on predefined criteria (Page et al., 2021). In this sense, the present study contributes to the literature by providing a structured, critical overview of recent empirical evidence on the academic, emotional, and physical impacts of excessive use of information and communication technology among university students. Notably, while previous work has explored positive links between physical activity and academic outcomes, for instance, improvements in attention, memory, classroom behaviour, and grades (Rasberry et al., 2011), such dimensions have rarely been considered in relation to digital overuse, despite their clear relevance to students' holistic development.

Previous studies have rarely considered such dimensions in relation to digital overuse Rasberry et al. (2011) rarely considered improvements in attention, memory, classroom behaviour, and grades in relation to digital overuse, despite their clear relevance to students' holistic development. These factors are fundamental, as they directly influence students' learning ability, motivation, and emotional well-being. Understanding these effects is crucial for designing educational and well-being strategies that mitigate the negative impacts of excessive use of digital technologies. Accordingly, the main objective of this systematic review is to analyse the combined effects of excessive video games and SM use on university students' mental well-being and AP. Specifically, it aims to (1) identify the academic and psychological consequences of excessive information and communication technology use among university students use, (2) explore gender and cultural patterns associated with these effects, and (3) highlight gaps and propose evidence-based recommendations for intervention and future research.

2. Materials and methods

The method employed an interpretive approach, using a systematic review of the scientific literature on the research topic, following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (Páramo, 2020). Articles and books on SMAs, physical and mental health, and the academic performance of university students were analyzed across Scopus, Web of Science, and Dialnet. Two reviewers independently assessed methodological quality and risk of bias using the Joanna Briggs Institute (JBI) Critical Appraisal Checklists (Munn et al., 2020), resolving discrepancies through discussion until they reached consensus. The procedure followed in this review consisted of extracting and analysing topics derived from the keywords of the research question, ensuring a rigorous collection and purification of information. The topics were extracted from the keywords derived from the research question to ensure a rigorous collection and purification of the information. The research question examined how excessive use of video games and social networks impacts university students' academic performance, mental well-being, and physical health.

To refine the search for these topics, an initial analysis of the most frequently used keywords in studies addressing this field was conducted. These "topics" represent the core dimensions analyzed in the review. These core dimensions were selected for their ability to capture the most significant aspects of digital overuse and its impact on students' holistic development. Focusing on these topics ensured that the review addressed both academic outcomes and psychological well-being, providing a comprehensive understanding of the issue. The purification process involved applying strict inclusion and exclusion criteria to select relevant studies (See Table 1), removing duplicates, and excluding articles that did not directly address the review objectives. From this stage, the most appropriate search equations were formulated by combining keywords using Boolean operators (AND, OR). To determine each study's affinity with the review's objectives, its relevance and appropriateness were assessed by reading the abstracts and, subsequently, the full texts.

The selection of databases and search terms was essential to ensure a comprehensive review of the impact of video games and social networks on the university students' physical, mental, and academic well-being. SCOPUS, Web of Science (WOS), and Dialnet were selected for their broad coverage and high-quality peer-reviewed publications. SCOPUS and WOS are known for their multidisciplinary focus and impact in social sciences, education, and health fields, central to this research in English. Dialnet enabled the inclusion of studies in Spanish, providing broader cultural perspectives. Even though the first studies go back to 2010, the final selected studies cover the period from 2019 to 2024 capturing recent technological advancements in SM and video game use and ensuring that both current impacts and recent interventions addressing problematic use are represented. Various search strategies were tested, initially retrieving 1, 247 documents, the search was subsequently refined to narrow down the results. The final search equation in English was: ("mental health" OR "well-being" OR "academic performance" OR "school performance") AND ("gaming disorder" OR "video game addiction" OR "social media addiction" OR "internet gaming disorder" OR "online gaming addiction") AND ("university students" OR "college students" OR "undergraduate students"). This yielded 159 papers in SCOPUS, 188 in WOS. For Spanish, the search equation was: ('mental health' OR 'well-being' OR 'academic performance') AND (addiction AND university AND social networks OR video games), and this yielded papers in Dialnet.

This review was conducted in accordance with the PRISMA 2020 guidelines for systematic reviews (Page et al., 2021). A total of 447 records were identified across the Scopus, Web of Science, and Dialnet databases; after removing 96 duplicates, 351 titles and abstracts were screened, resulting in 188 full-text articles assessed for eligibility and 16 studies included in the final synthesis. The screening and selection were carried out independently by two reviewers, with discrepancies resolved

Table 1
Criteria used for inclusion and exclusion of studies.

Inclusion criteria	Exclusion criteria
- Studies published in English or Spanish. -Studies addressing the relationship between SM, video games, and AP in university students. -Studies published at any time. -Studies include analyses on mental health, physical health, and AP. -Research conducted on university students around the world.	-Studies published in languages other than English or Spanish. -Studies that do not analyse the impact of SM or video games in the context of university education -Studies that do not present mental, physical, or AP. -Studies focusing on another educational stage.

by discussion and consensus with a third reviewer. The study selection process is illustrated in the PRISMA flow diagram (Fig. 1), including reasons for exclusion. Quality appraisal was performed using the Joanna Briggs Institute (JBI) critical appraisal checklists for each study design, and studies were classified into three quality levels (low, moderate, high). Only studies scoring above 60 % were retained in the final synthesis to ensure methodological rigor and reliability of the results.

Data extraction was conducted using a standardized spreadsheet capturing author, year, country, sample size, methods, key findings, and relevance to AP and/or mental well-being. A thematic synthesis approach was employed to identify patterns across studies and to organize them according to three analytic dimensions: AP, mental health outcomes, and excessive use of information and communication technology. The analysis of the data included in this review was conducted through an interpretative thematic synthesis approach. This method allows for the identification, organization, and in-depth analysis of recurring patterns within the findings of the selected studies, facilitating a comprehensive and contextualized understanding of the relationships between excessive video games and social media use and their effects on academic performance, mental, and physical health among university students. A systematic process was followed to code relevant results from each study. This approach is especially suitable for studies with heterogeneous results, as those included in this review, and contributes to integrating both quantitative and qualitative evidence within a comprehensive and critical framework. This methodology has allowed us to identify gaps and future directions for research and intervention in this field.

3. Results

The results section synthesizes the findings of 16 studies, focusing on the relationship between excessive use of video games and social networks and academic performance, mental health and physical well-being of university students, which have been conducted in different countries. The selected research studies were published in English (100 %; N = 16). All publications were sought although the selected studies were written between 2019 and 2024, ensuring up-to-date results on behavioural addictions and AP. The studies were conducted in 6 Asian countries (Saudi Arabia, Bangladesh, China, Pakistan, Lebanon, and Turkey 56.25 %), 4 African countries (Botswana, Ethiopia, Namibia, South Africa 25 %), 2 European countries (United Kingdom and Romania 12.50 %), and 1 American country (Honduras 6.25 %) with Bangladesh conducting the most studies (3), followed by China, Turkey and Ethiopia with 2 studies each country. 15 of the studies used quantitative methodology (93.75 %; N = 15), and 1 used a qualitative methodology through a systematic literature review (6.25 %; N=1). Among the instruments or study used documentation, 47 (questionnaires and scales) were used, of which a total of 8 instruments matched in the studies were the Bergen Social Media Addiction Scale (n = 7), Work engagement scale students (n = 3), Patient health questionary (n = 3), SMAS-SF (n = 2), GAD-7 (n = 2), BDI (n = 2), G-20 (n = 2), and STAI (n = 2). Also, the students' ratings were taken into account. Table 2 provides all the information, including authorship, year of publication, purpose, sample, country, methodology with assessment instrument, and main results.

3.1. Assessment of quality and bias in the included studies

To assess the risk of bias and methodological quality of the included studies, the Joanna Briggs Institute (JBI) critical appraisal checklists were used, adapted according to the study design. Two different checklists were applied: one for cross-sectional quantitative studies (n =

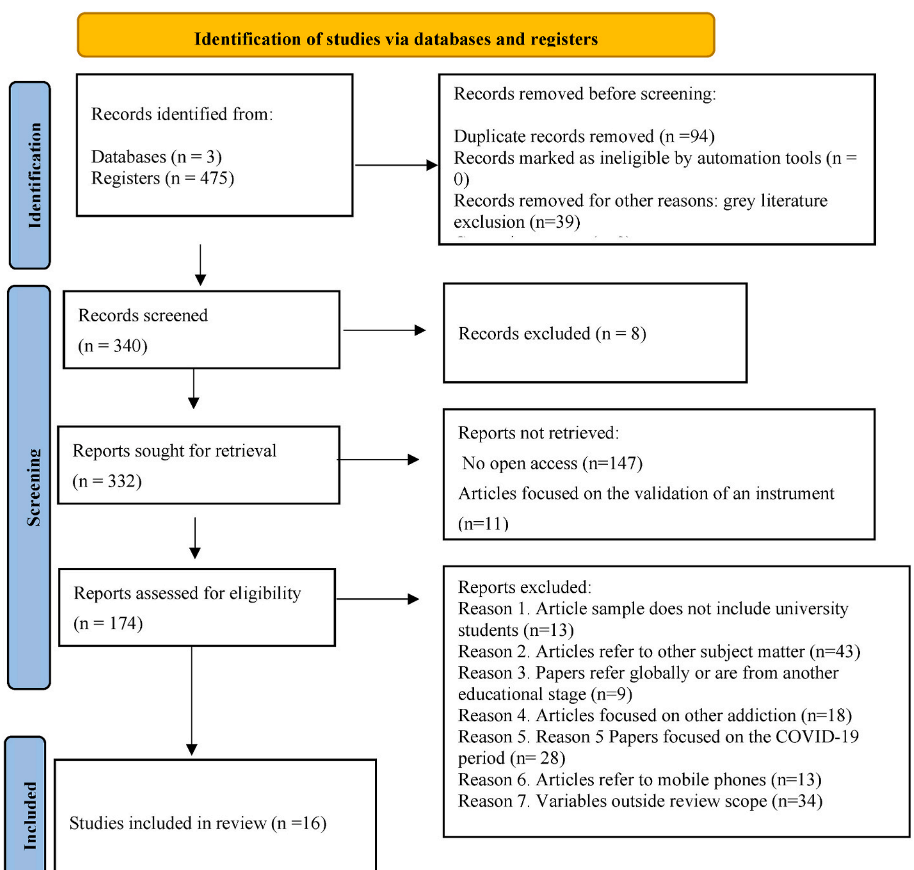

Fig. 1. PRISMA flowchart of literature search and study selection.

Table 2
Summary of the main characteristics of the selected studies.

Author/Year	Objectives	Sample	Country	Methodology	Results
Alheneidi, y Smith, (2021).	To analyse cross-sectional associations by examining the prevalence of information overload, internet addiction and SMA. Assess their effects on students' general well-being and academic outcomes.	226 university students	United Kingdom	Quantitative, with wellbeing questionnaires and an online survey. The following were used to assess experiences with SMA: The Overload Scale, The Internet Addiction Test, The Student WPQ (multidimensional well-being scale), and The Bergen Social Media Addiction Scale (BSMAS).	Internet addiction increased academic stress and was negatively correlated with smoking and sleep quality. Only information overload predicted negative evaluations. Problematic social network use was an independent predictor of AP, with the best students using more social networking applications.
Aslan y Polat (2024)	This study aims to determine the level of SMA, its impact on academic success, and the interactions between SMA, loneliness, depression, life satisfaction, problem-solving skills, and academic self-efficacy factors.	490 undergraduate nursing student	Turkey	Quantitative with questionnaires: Social Media Addiction Scale-Student Form (SMAS-SF) to measure the level of SMA, Beck Depression Inventory (BDI): It provides scores that classify participants into different levels of depression. State-Trait Anxiety Inventory (STAI) assesses both temporary (state) and long-term (trait) anxiety in participants, providing information about their general and situation-specific anxiety levels.	The study found moderate SMA, mild loneliness, and moderate depression among the students. Despite having high academic self-efficacy and good problem-solving skills, they showed moderate GPA and mental health problems. Students who visited SM more than 15 times daily had the lowest satisfaction. Females were more depressed, but males were more addicted and lonelier. Living with friends increased addiction. University students were spending more than 7 h per day on social networks and accessing them more than 15 times a day daily — behaviours that are considered indicators of SMA — have been shown to negatively impact AP.
Chowdhury et al. (2024)	Finding risk factors and predicting anxiety, depression, and insomnia among university students in Bangladesh using tree-based ML models.	1250 university students	Bangladesh	Quantitative with questionnaires: The Bergen Social Media Addiction Scale (BSMAS), Generalised Anxiety Disorder-7 (GAD-7), to assess anxiety. Patient Health Questionnaire-9 (PHQ-9) to measure depression., Insomnia Severity Index (ISI-7) to evaluate insomnia problems, and the Spanish version of the reduced scale of the Morningness-Eveningness Questionnaire (rMEQ) to observe unique variations in nocturnal preferences, sleep-wake activity patterns and attention in the morning and evening. These data were processed using tree-based machine learning models such as XGBoost to predict the disorders above and the Insomnia Severity Index (ISI-7).	It was found that university students in Bangladesh faced high levels of anxiety, depression, and insomnia, with key risk factors including SMA, age, AP, smoking, family income, and sleep patterns.
David et al. (2024)	To assess social networking addiction in Romanian undergraduate nursing students and its association with AP, depression, and anxiety.	90 university students	Romania	A quantitative survey using the Romanian Social Media Addiction Scale-Student Form (SMAS-SF), the Beck Depression Inventory (BDI), and the State-Trait Anxiety Inventory (STAI), which uses scales measuring temporary anxiety known as state anxiety (A-State), and long-term anxiety known as trait anxiety (A-Trait).	Mental disorders negatively affected AP, highlighting that social network addiction and academic engagement mediated the relationship between social anxiety and AP. However, the impact of social anxiety was more related to academic engagement than to network addiction.
Dule et al. (2023)	To assess Facebook addiction and its relationship with AP and self-esteem, anxiety and depression symptoms, and study habits in the university students. .	422 university students	Ethiopia	Quantitative Facebook addiction was examined with the Bergen Facebook Addiction Scale (BFAS). The Rosenberg Self-Esteem Scale (RSES), the Hospital Anxiety and Depression Scale (HADS), and the Study Habits Questionnaire (SHQ) were used to respectively assess self-esteem, anxiety and	Facebook addiction was observed more frequently in the study participants, which was detrimental to their AP. It was also associated with impaired mental well-being and decreased self-esteem.

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Table 2 (*continued*)

Author/Year	Objectives	Sample	Country	Methodology	Results
				depression symptoms and study habits.	
Hawi y Samaya (2024)	To find out the relationship between gambling disorder, attention deficit hyperactivity disorder (ADHD), and grade point average in the university students.	348 university students whose mean age was 19,93, *SD* = 2,08.	Lebanon	Quantitative and survey. The IGD-20 test for gambling addiction was administered. ADHD symptoms were assessed using the Adult ADHD Self-Report Scale (ASRS v1.1)	There is a significant positive relationship between gambling addiction and ADHD, primarily due to shared traits like impulsivity, especially in females. Both disorders negatively impact AP, particularly in high-achieving students, suggesting that their symptoms may hinder success.
Kassaw y Demareva (2023)	To identify the main factors that impede the performance of the Ethiopian university students.	27 scientific papers on Ethiopian university students.	Ethiopia	Qualitative with a theoretical approach through a systematic review of the literature.	Inadequate classroom environments, dysmenorrhea, and excessive SM use are associated with poor AP. In contrast, healthy sleep habits, high entrance exam scores, and avoiding substance abuse contribute to academic success, highlighting the links between students' mental and physical well-being and academic outcomes.
Landa-Blanco et al. (2024)	To analyse how SMA is related to academic engagement in university students, considering the mediating role of self-esteem, depressive symptoms and anxiety.	412 students enrolled at the National Autonomous University of Honduras.	Honduras	Quantitative with a non-experimental-relational design. A set of questionnaires comprising Utrecht Work Engagement Scale-Student Version, Bergen Social Media Addiction Scale Patient Health Questionnaire, generalised Anxiety Disorder (GAD-7), Rosenberg Self-Esteem Scale and a questionnaire with general student data were applied.	A non-linear relationship between SM use and AP was identified, highlighting the importance of considering mediating factors. SM use was found to negatively affect mental health and productivity by lowering self-esteem and increasing depressive symptoms, which, in turn, contributed to procrastination and disengagement.
Le Roux et al. (2021)	This study assesses whether online vigilance predicts variations in the university students AP, considering general media use and multitasking behaviour as a control variable.	1445 university students	South Africa, Botswana and Namibia	The quantitative survey and Online Vigilance Scale measure individuals' online vigilance and attentiveness to assess how people handle information in a digital environment, their ability to concentrate and stay alert in the face of distractions, and how well they are able to monitor their online activities. The Media Multitasking Index-Short (MMI-S) assesses people's tendency to multitask with different media.	There was evidence that the general use of internet-based media may not harm university students. When controlling for media use behaviours, online surveillance was a stronger predictor of AP.
Mahmud et al. (2023)	To determine the prevalence of online gaming addiction among Bangladeshi university students and how it affected both their interpersonal relationships and AP.	399 university students	Bangladesh	Quantitative with an ad hoc questionnaire created with Google to assess variables such as the number of hours spent gambling, addiction tendencies, and AP.	Students played online more than 30 h per week; males were more likely to show symptoms of addiction. Age, internet access, and gaming frequency influenced this trend. Excessive gaming was associated with a lower grade point average, less physical activity, and less study time, negatively affecting AP.
Mosharrafa et al. (2024)	To assess university students' use of social networking sites and their impact on AP and mental health	360 university students, 60 % men and 40 % women. El 61, 6 % among 21–26 years old	Bangladesh	Quantitative. Measurement scales Bergen Social Media Addiction Scale	Using social networking sites improved students' AP by positively influencing their psychological well-being, with mental health mediating between SMA and academic success.
Mou et al. (2024)	To assess how social anxiety and academic engagement influence AP, and to analyse the mediating role of SMA in this relationship.	2661 university students	China	Quantitative with a self-report questionnaire including the Liebowitz Social Anxiety Scale, the Bergen SSR Addiction Scale, the Utrecht Student Work Engagement Scale, and GPA.	Social anxiety negatively affected AP, mediated by academic engagement and SMA.
Nasrullah y Khan (2019).	To determine whether there is a positive impact of the use of SM on the AP of the university students, as well as on their	64 university students from the Community College, the Faculty of Business Administration, the Faculty of	Saudi Arabia	Quantitative with Ad hoc questionnaire derived from the research questionnaire used in Oman. (El Khatib & Khan, 2017).	Students favoured making friends on SM over in-person interactions, finding it addictive yet enjoyable. While not ideal

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Table 2 (*continued*)

Author/Year	Objectives	Sample	Country	Methodology	Results
	interpersonal and social skills at Prince Sattam bin Abdulaziz University.	Education, the Faculty of Engineering and the Faculty of Pharmacy.			for learning, these platforms helped exchange study materials and updates. Many believed they improved communication skills, with some teachers promoting academic use only. A link was suggested between compulsive social media use, emotional well-being, and AP.
Pekpazar et al. (2021)	We are measuring the effects of self-esteem on Instagram addiction, procrastination, and AP.	378 university students.	Turkey	Quantitative with the survey as the instrument. Kircaburun and Griffith's Instagram Addiction Scale (IAS), the Turkish version of Tuckman's procrastination scale, and the Rosenberg Self-Esteem Scale developed by Rosenberg were used. GPA with grade point average was used.	Instagram addiction significantly increased procrastination and negatively impacted AP. Students addicted to Instagram often delayed their homework, which affected their grades. As self-esteem declined, both addiction and procrastination rose. While Instagram addiction and self-esteem influenced procrastination, their direct effect on AP was minimal.
Zahra et al. (2020)	Establishing the relationship between online gambling and emotional intelligence, psychological distress, and AP among college students also investigated whether playing at different times could influence psychological distress and emotional intelligence.	315 university students aged 18–25 years ($M = 23.97$, $SD = 6.07$).	Pakistan	Quantitative. Instruments: Internet Gambling Disorder Test (IGD-20 Test; Pontes et al., 2014).Wong and Law Emotional Intelligence Scale (WLEIS; Wong & Law, 2002), Depression Anxiety and Stress Scale (DASS-21; Lovibond & Lovibond, 1995). Grades of the university students were also considered.	Internet gaming was positively associated with psychological distress and negatively correlated with both emotional intelligence and AP. Students who played late at night, especially after midnight, showed higher distress levels. Overall, increased gaming time corresponded to greater distress and lower AP, with no significant link to emotional intelligence.
Zhuang et al. (2023)	To evaluate a serial mediation model that relates SMA to the academic commitment of the university students, sleep quality and fatigue, both linked to psychosomatic health, are considered mediating variables.	2.661 university students (43.3 % male, mean age = 19.97).	China	Quantitative with the survey. Participants completed the Bergen Social Media Addiction Scale, Utrecht Work Engagement Scale-Student Version, Pittsburgh Sleep Quality Index (PSQI), Social Media Addiction Scale (SMAS), Fatigue Severity Scale (FSS), and Academic Engagement Scale.	SMA in the university students was negatively related to academic engagement (Effect = −0.051). Sleep quality and fatigue mediated this relationship, with mediation effects of −0.031 and −0.109, respectively. The serial effect of both variables was −0.080, and the total indirect effect reached 80.9 %.

15) and another for the qualitative study (n = 1). These tools evaluate key aspects such as clarity of inclusion criteria, validity of measurement tools, appropriateness of statistical analyses, and transparency in reporting results. The systematic application of these checklists allowed for the identification of relevant methodological strengths and weaknesses, as well as potential sources of bias, in the reviewed studies.

3.1.1. *Analytical cross-sectional studies*

Most of the 16 cross-sectional studies demonstrated rigorous methodology across most domains. All studies clearly defined their inclusion criteria and provided detailed descriptions of the study subjects and settings, enhancing transparency and replicability. Similarly, exposure and outcomes were generally measured using valid and reliable tools. However, identifying and managing confounding factors is a recurrent source of potential bias. Specifically, four studies (Alheneidi & Smith, 2021; David et al., 2024; Landa-Blanco et al., 2024; Mosharrafa et al., 2024) did not report whether confounding variables were identified or strategies were implemented to control them, suggesting a potential risk of internal validity issues. These omissions may affect the accuracy of causal interpretations. On the other hand, all studies used appropriate statistical analysis, supporting the robustness of their conclusions regarding associations between exposures and outcomes.

3.1.2. *Qualitative research*

The qualitative study by Kassaw and Demareva (2023) was evaluated using the JBI checklist for qualitative research. The study showed a high degree of methodological congruence regarding its philosophical perspective, methodology, and data collection, representation, and interpretation. However, there are some areas of concern regarding researcher reflexivity and ethical transparency.

- The influence of the researchers on the research process was not addressed, and the researchers' cultural or theoretical positioning was unclear.
- Additionally, there was insufficient information on ethical approval, which raises potential concerns about the ethical integrity of the study.
- Participants' voices were not represented, limiting insight into the authenticity of the data.

Despite these limitations, the overall appraisal still judged the study to be of acceptable quality, though caution should be exercised when interpreting its findings. This table presents the methodological appraisal of studies using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist. The assessment allows readers to critically evaluate the strengths and limitations of the included studies, bringing transparency and rigour to the systematic review process. It also highlights methodological gaps that need to be addressed in future research, ensuring that subsequent studies are ethically sound.

This table presents the methodological appraisal of studies using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Qualitative Research. It highlights the strengths and limitations of each study with respect to philosophical alignment, data collection, analysis, and ethical considerations. By providing this detailed assessment, the table enables readers to critically evaluate the quality of the evidence and identify areas for improving future qualitative research, particularly in terms of reflexivity and transparency.

It is important to provide a summary of the overall reliability and methodological quality of the included studies based on the appraisal checklists presented in Tables 3 and 4. Most of the quantitative studies (10 out of 15) analyzed in this review demonstrate adequate methodological quality, with explicit inclusion and exclusion criteria, reliable measurement tools, and appropriate statistical analyses, which contribute to the overall robustness of the results. However, in four studies, the identification and control of confounding factors were not sufficiently clear, which could lead to bias and affect internal validity and causal conclusions. Due to this limitation, although there is a consistent relationship between excessive use of digital media and adverse effects on mental health and academic performance, interpreting the strength and direction of this relationship should be done with caution. Furthermore, the observed shortcomings in participant representation and the lack of ethical clarity in the one qualitative study reinforce the need for caution in extrapolating these results. Taken together, these issues highlight the importance of future research that employs more effective strategies to detect and control for confounding variables and that ensures greater transparency and ethical rigor to reinforce the validity and relevance of the results.

As shown in Fig. 2, the production of scientific literature on this topic increased significantly in 2021, during the pandemic, suggesting a surge in research interest and scholarly activity driven by the global health crisis. However, this upward trend did not continue into 2022, where the number of publications remained relatively stable or even experienced a slight decline. Interestingly, from 2023 onwards, there has been a marked resurgence in publication output, with a considerable increase observed in both 2023 and 2024, indicating renewed or sustained interest in the field and possibly reflecting the ongoing evolution of research priorities and emerging scientific challenges.

As illustrated in Fig. 3, most of the countries included in the analysis contributed to only one study, reflecting relatively limited participation in research on this topic. However, there are notable exceptions: Bangladesh stands out for its participation in three different studies, suggesting a higher level of research activity or interest in this area. Similarly, China, Ethiopia, and Turkey each contributed to two studies, indicating moderate but significant participation compared to other countries. These variations in research output may reflect differences in national research priorities, available resources, or the local relevance of the topic. The reviewed studies reveal several ways in which the problematic use of the internet, social networks, and video games impacts the mental, physical, and academic well-being of university students. To better organize and compare findings, the studies are grouped into five key dimensions: 1) impact on AP, 2) gender differences and mental well-being, 3) physical health, 4) sleep quality and academic engagement, and 5) intervention proposals. These dimensions capture the main themes, facilitating understanding of the interconnected impacts.

3.2. *Academic impact*

Studies by Alheneidi and Smith (2021), Dule et al. (2023), Mahmud et al. (2023), Nasrullah and Khan (2019), Pekpazar et al. (2021), and Zahra et al. (2020) indicate that problematic SM and video game use negatively correlate with AP. Mahmud et al. (2023) found that students

Table 3
Joanna Briggs Institute (JBI) critical appraisal checklist.

N°	Question/ Study	1. Were the criteria for inclusion in the sample clearly defined?	2. Were the study subjects and the setting described in detail?	3. Was the exposure measured in a valid and reliable way?	4. Were objective, standard criteria used for measurement of the condition?	5.Were confounding factors identified?	6. Were strategies to deal with confounding factors stated?	7. Were the outcomes measured in a valid and reliable way?	8. Was appropriate statistical analysis used?
1	Alheneidi and Smith (2021).	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes	Yes
2	Aslan and Polat (2024)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3	Chowdhury et al. (2024)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
4	David et al. (2024)	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes	Yes
5	Dule et al. (2023)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
6	Hawi and Samaha (2024)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
8	Landa-Blanco et al. (2024)	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes	Yes
9	Le Roux et al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
10	Mahmud et al. (2023)	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes	Yes
11	Mosharrafa et al. (2024)	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes	Yes
12	Mou et al. (2024)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
13	Nasrullah and Khan (2019).	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
14	Pekpazar et al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
15	Zahra et al. (2020)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
16	Zhuang et al. (2023)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 4
JBI critical appraisal checklist for qualitative research.

N°	Question/ Study	1. Is there congruity between the stated philosophical perspective and the research methodology?	2 Is there congruity between the research methodology and the research question or objectives?	3. Is there congruity between the research methodology and the methods used to collect data?	4. Is there congruity between the research methodology and the representation and analysis of data?	5. Is there congruity between the research methodology and the interpretation of results?	6. Is there a statement locating the researcher culturally or theoretically?	7. Is the influence of the researcher on the research, and vice versa, addressed?	8. Are participants, and their voices, adequately represented?	9. Is the research ethical according to current criteria, and is there evidence of ethical approval?	10. Overall quality
1	Kassaw and Demareva (2023)	Yes	Yes	Yes	Yes	Yes	Unclear	No	Unclear	Unclear	Yes

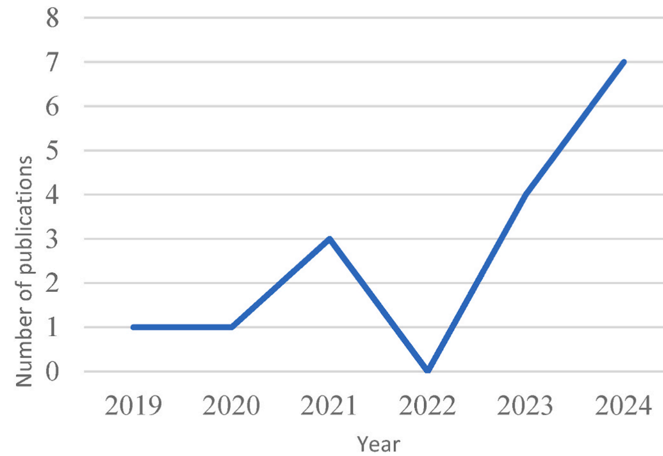

Fig. 2. Number of publications per year among the selected studies.

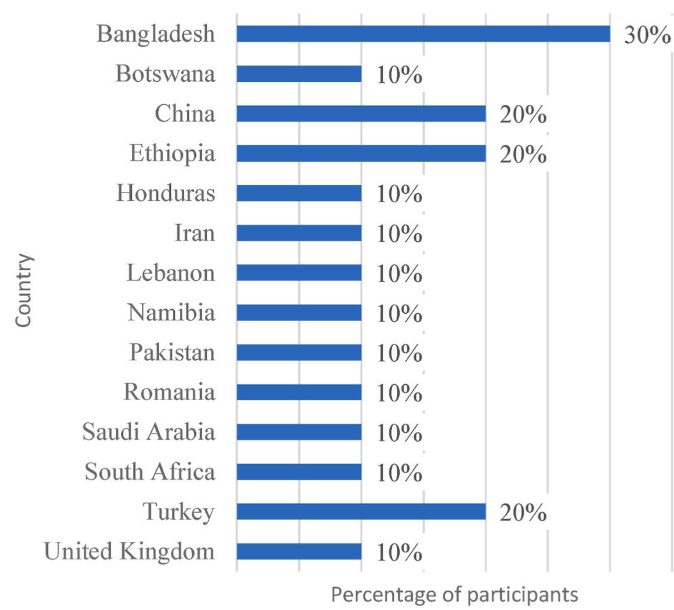

Fig. 3. Distribution of participants across the countries included in the study.

who spent around 30 h per week on video games had poor AP, consistent with Pekpazar et al. (2021), who linked Instagram addiction to procrastination, negatively impacting AP. Alheneidi and Smith (2021) observed that while SM use does not always directly impact AP, information overload significantly predicts negative evaluations. Le Roux et al. (2021) noted the complex, non-linear relationship between SM use and AP, influenced by factors like the need to stay updated on online platforms. Moderating factors such as time management, concentration, and study skills were essential to buffer the adverse effects of SM use on AP. Of the 16 studies analyzed, 7 studies (43.75 %) addressed the impact of excessive SM and video game use on AP. These studies consistently suggest a negative association between problematic digital consumption and students' academic outcomes. Furthermore, these findings underline the importance of monitoring and managing students' digital habits, as excessive use of social media and video games can significantly hinder academic performance, highlighting the need for tailored interventions and educational strategies to mitigate these negative effects.

3.3. *Gender differences and mental well-being*

Studies by Aslan and Polat (2024), Hawi and Samaha (2024), and

Moreno (2019) highlighted gender-based differences, with women being more vulnerable to SM and video game addiction's psychological impacts, such as depression and anxiety. Aslan and Polat (2024) found higher life satisfaction among men, while women were more prone to depression, consistent with Hawi and Samaha (2024), who identified a more vital link between video game addiction and ADHD symptoms in women. Moreno (2019) and Lopez (2022) emphasized Instagram's adverse effects on women's mental health, impacting AP. Nasrullah and Khan (2019) suggested that social networks could enhance academic collaboration and peer communication well-being. Of the 16 studies reviewed, 5 studies (31.25 %) explicitly addressed gender differences and/or impacts on mental well-being related to SM and video game overuse. Overall, the findings underscore that women are disproportionately affected by the negative psychological consequences of digital overuse, particularly through platforms like Instagram. In contrast, men tend to report higher life satisfaction, although they are not immune to other risks like addiction or reduced AP.

3.4. Mental health outcomes

An innovative approach by Chowdhury et al. (2024) employed machine learning (XGBoost) to diagnose mental health issues stemming from SM use, identifying problems like anxiety and depression more effectively than traditional methods. David et al. (2024) and Mou et al. (2024) recommend combining psychological interventions to counter SMA's effects, and both agree that early, personalized interventions improve students' emotional well-being. Out of the 16 reviewed studies, 3 studies (18.75 %) specifically focused on innovative diagnostic tools and intervention strategies to address mental health problems resulting from excessive SM use. The available evidence reveals an emerging trend towards the use of advanced technologies for early diagnosis, as well as a consensus on the value of personalized psychological interventions as key strategies to prevent or counteract the psychological impact of excessive SM and video game use in higher education settings.

3.5. Physical health effects

Studies such as Mahmud et al. (2023) link SM and video game use to sedentary lifestyles and related health problems, with prolonged screen time associated with symptoms like headaches and reduced quality of life. In contrast, Mosharrafa et al. (2024) proposed that, in specific contexts, SM may positively affect mental health, suggesting that the context and type of SM use are critical to determining health outcomes. Among the 16 studies reviewed, 2 studies (12.5 %) addressed the physical health consequences of excessive SM and video game use. Although the literature directly addressing the physical health effects is limited, findings suggest a clear association between screen overuse and health risks, highlighting the complex and context-dependent nature of these outcomes. These findings underscore the need for more comprehensive research on the implications for physical health, as well as the development of interventions that balance the potential benefits and risks of SM and video games.

3.6. Sleep quality and academic engagement

Sleep quality emerged as a crucial factor in students' academic engagement and general well-being. Kassaw and Demareva (2023) and Zhuang et al. (2023) found that poor sleep quality exacerbated reduced academic engagement in students with SMAs, highlighting the need to promote healthy sleep habits to improve AP and physical well-being. This suggests that the intersection of sleep, technology addiction, and AP warrants further exploration. 2 out of 16 studies (12.5 %) explicitly examined the relationship between sleep quality, academic engagement, and excessive SM use among university students. These findings underscore the importance of healthy sleep routines in mitigating the adverse academic and health consequences of excessive screen time. Moreover, they suggest that sleep quality is a key mediating factor in the relationship between technology use and students' academic and personal functioning.

3.7. Intervention strategies

David et al. (2024) and Mou et al. (2024) provide insights on how interventions, including cognitive-behavioral strategies, mentoring, and academic support, help mitigate SMA's adverse effects and improve AP. They agree on the importance of academic engagement in countering SM's negative impact on mental and academic health, and suggest that educational institutions develop programs to enhance students' overall well-being. Chowdhury et al. (2024) further proposed leveraging advanced technologies, such as machine learning, to detect mental health issues early, enabling more tailored interventions. 3 out of 16 studies (18.75 %) proposed or evaluated intervention strategies to address the adverse effects of SMA and excessive video game use on students' AP and mental health. These studies suggest that integrating psychological, technological, and academic support can be a promising path forward for universities seeking to address the growing issue of digital addiction among students. Collectively, these studies suggest that targeted interventions can play a key role in supporting students' mental health and academic outcomes. These findings highlight the detrimental effects of excessive SM and video game use on self-esteem and mental well-being, suggesting that promoting healthier online habits and addressing underlying self-esteem issues could mitigate these risks.

Studies by Dule et al. (2023), Landa-Blanco et al. (2024), and Pekpazar et al. (2021) emphasize the significant relationships between problematic internet use, SM, and video games and the mental, physical, and academic well-being of university students. Information overload and SMA have consistently shown negative impacts, particularly on mental health and AP, with studies such as Alheneidi and Smith (2021) indicating information overload as a critical predictor of poor academic evaluations. Studies by Aslan and Polat (2024), Moreno (2019), and Hawi and Samaha (2024) noted that SM and video games most profoundly affect women, who are more prone to depression and negative emotional impacts. Additionally, the correlation between SMA and ADHD symptoms, especially among females, reflects this group's heightened susceptibility to technology's adverse effects on AP.

Regarding physical health, studies such as Mahmud et al. (2023) confirm that prolonged SM use is associated with sedentary lifestyles and related health risks. However, Mosharrafa et al. (2024) propose that SM can positively impact mental well-being in specific contexts, suggesting that usage type and context moderate adverse effects. Sleep quality also emerged as critical; Zhuang et al. (2023) found that poor sleep exacerbated the adverse effects of SMA on AP, underscoring the importance of promoting healthy sleep habits among students. Overall, the findings of this review reveal consistent patterns associating excessive digital consumption with impairment in both mental health and academic performance, highlighting the urgency for interventions tailored to the university context.

4. Intervention proposals

Combined strategies are recommended, including psychological support, mentoring, and responsible technology use education. David et al. (2024) and Mou et al. (2024) advocate for programs that encourage academic engagement and emotional management support to reduce the adverse effects of SM and video game use. Chowdhury et al. (2024) suggested advanced technologies like machine learning to detect early mental health issues tied to technology use, allowing for more personalized interventions. The results collectively indicate that SM and video game addiction not only affect AP but also compromise the US's mental and physical health. However, usage type and context play a moderating role, underscoring the potential for responsible technology use to mitigate adverse effects. These findings support future research and public health strategies encouraging balanced technology use to enhance students' academic and personal well-being.

5. Bias analysis

5.1. Biases in the included studies (narrative synthesis based on results)

Most of the 16 included studies demonstrated generally strong methodological quality, with clearly defined inclusion criteria and valid measurement tools. Nevertheless, some potential biases were identified. Among the 15 quantitative cross-sectional studies, a key limitation was the inadequate identification and control of confounding factors in four studies, which may affect the internal validity and causal interpretations. Despite this, all quantitative studies applied appropriate statistical analyses supporting their findings. The single qualitative study demonstrated overall methodological coherence but raised concerns regarding researcher reflexivity, ethical transparency, and the representation of participants' perspectives. These issues could impact on the authenticity and ethical rigor of the findings. Moreover, while the studies provide valuable insights, these methodological limitations highlight the need for cautious interpretation of results.

5.2. Biases in the present systematic review

This review faces several potential sources of bias that should be considered when interpreting the findings. Restricting the selection to studies published in English and Spanish may have excluded relevant research in other languages, and the focus on recent publications may limit the identification of longer-term trends. Moreover, most included studies relied on self-reported questionnaires, increasing the risk of response bias, and variations in how social media and video game addiction were defined across studies introduce inconsistencies in measurement and interpretation. The geographical concentration of research in Asian and African university contexts also limits the generalizability of the results to other cultural environments. Despite these limitations, the studies reviewed provide coherent and valuable insights into how problematic social media and video game use may influence university students' well-being and academic performance, underscoring the need for future research that expands methodological rigor, cultural representation, and conceptual clarity.

6. Discussion

The results of this review indicate a consistent association between excessive use of social media and video games and decreased academic performance, as well as negative effects on psychological and physical well-being among university students. These patterns must be understood in relation to broader sociocultural transformations, including the rapid digitization initiated in the 1990s, the expansion of large-scale social networking platforms since the mid-2000s, and particularly the intensification of digital dependence during the COVID-19 pandemic, which expanded the use of digital technologies for academic, social, and recreational purposes. Although the studies converge on general tendencies, such as reduced academic performance and increased emotional vulnerability, heterogeneity in cultural contexts, methodological designs, and definitions of addiction prevents absolute generalization. Therefore, the interpretation of these findings should consider both statistical evidence and the social and psychological environments in which students engage with digital platforms.

Consistent trends across studies show that problematic social media use is associated with increased stress, anxiety, depression, and social isolation (Zahra et al., 2020; Alheneidi & Smith, 2021; Kassaw & Demareva, 2023; Landa-Blanco et al., 2024; Mou et al., 2024; Lozano-Blasco & Cortés-Pascual, 2020; Valencia-Ortiz et al., 2021). Moreno (2019) and López (2022) identify Instagram as particularly influential in promoting appearance-based comparison and anxiety, especially among young women. Varchetta et al. (2020) found that excessive social media use heightens fear of social exclusion, while Noorunnahar et al. (2023), Aslan and Polat (2024), and Chowdhury et al. (2024) confirm an inverse association between social media addiction and academic performance. Gender patterns emerge as a relevant dimension: Valencia-Ortiz et al. (2021) and Alarcón-Allaín and Salas-Blas (2022) report reductions in emotional intelligence associated with increased addiction, particularly among women. Meanwhile, Mahmud et al. (2023) found that male students tend to show greater vulnerability to video game addiction, suggesting differentiated psychological and motivational mechanisms depending on platform type (Pérez-Jorge & Martínez-Murciano, 2022).

Further research highlights the role of platform-specific effects. Data from We Are Social & Hootsuite (2022) indicate higher social media usage among women aged 19–24. Hawi and Samaha (2024) identify a higher female tendency toward video game addiction associated with symptoms of ADHD, while Mahmud et al. (2023) report higher gaming addiction rates in males, illustrating complex gender-specific patterns. Regarding platform type, Dule et al. (2023) confirm that Facebook addiction negatively affects emotional well-being, while Pekpazar et al. (2021) show that lower self-esteem increases vulnerability to Instagram addiction, which contributes to procrastination. Mahmud et al. (2023) report university students spending over 4 h daily on social media, exceeding the average reported by Dzul and Cauich (2021, pp. 87–109). Nasrullah and Khan (2024) suggest that the preference for online interaction over face-to-face engagement may contribute to the persistence of excessive use.

Video game overuse similarly affects academic and physical functioning. Mahmud et al. (2023), Zahra et al. (2020), Alzahrani and Griffiths (2024), and Barreto-Cabrera et al. (2024) confirm that prolonged gaming reduces physical activity and is associated with poorer academic performance. Physical health variables such as sleep quality, dysmenorrhoea, and smoking also influence academic engagement (Alheneidi & Smith, 2021; Kassaw & Demareva, 2023; Zhuang et al., 2023). High-performing students appear to use social media selectively. In contrast, lower-performing students often report disrupted sleep and concentration due to excessive use (Alhusban et al., 2022), reinforcing the need to regulate digital habits to protect cognitive performance.

Individual differences, including self-regulation skills, coping strategies, personality traits, and academic pressures, moderate these effects. Therefore, interventions must extend beyond technological restrictions to include socio-emotional skill development. Universities are uniquely positioned to promote responsible digital habits by integrating media literacy, emotional regulation, time management, and resilience training into academic support programs.

In summary, excessive use of social media and video games is consistently associated with declines in academic performance and student well-being, although the extent of these effects varies depending on emotional, physical, and behavioural factors. As platforms evolve and incorporate increasingly immersive and attention-demanding features, future research must continue to examine these emerging dynamics. Educational and mental health policies should prioritize promoting balanced digital behaviours and developing adaptive strategies that reflect the changing technological landscape.

7. Conclusions

This study examined how addiction to video games and social networks affects the physical, mental, and academic well-being of university students. The findings consistently indicate that problematic use of these technologies negatively impacts academic performance (AP), mental health, and physical health. Regarding AP, Mahmud et al. (2023) and Pekpazar et al. (2021) showed that excessive engagement with video games and social networks increases procrastination, leading to poorer academic outcomes. Alheneidi and Smith (2021) reinforce this by demonstrating that information overload contributes to negative academic evaluations. In terms of mental health, research suggests a greater vulnerability among women, with Aslan and Polat (2024) and Hawi and Samaha (2024) reporting higher levels of depressive symptoms and ADHD traits associated with problematic use, consistent with Moreno's (2019) observations about Instagram's emotional impact. Additionally, excessive video game use is linked to physical health problems, including headaches and dizziness derived from sedentary habits (Mahmud et al., 2023).

Importantly, recent studies highlight potential strategies to mitigate these adverse effects. David et al. (2024) and Mou et al. (2024) suggest that combined interventions, such as psychological support, mentoring, and structured academic guidance, are effective in reducing the negative consequences of addiction to digital platforms. Chowdhury et al. (2024) emphasize the value of emerging technologies for detecting and addressing mental health issues associated with excessive use at an early stage. These findings underscore the need for universities and health professionals to adopt proactive and adaptive approaches as digital behaviours continue to evolve.

In summary, addiction to social networks and video games detrimentally affects students' academic performance and mental and physical well-being. Educational institutions should prioritize evidence-based intervention strategies that promote balanced technology use and healthier daily habits while strengthening students' emotional regulation and time management skills. Future research should explore how contextual differences influence patterns of digital engagement and focus on the development of personalized prevention and support programs.

CRediT authorship contribution statement

María Carmen Martínez-Murciano: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Conceptualization. **David Pérez-Jorge:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of use of IA

During the preparation of this work, the authors used ChatGPT 4.0 to improve the clarity and fluency of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and took full responsibility for the publication's content.

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Data availability

This study is a systematic review based on publicly available data from previously published articles. The search strategy and inclusion/exclusion criteria are detailed in the methodology section. No new data were generated during this study.

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